STM32L433 IC4 <-> TB6752249 IC2 - stm32l433_tb67s249_connection.md

AGC straps: R50=0; R52/R54/R56=100k; R51/R53/R55/R57=DNP.
CLIM1/BOOST/FLIM pull-ups must use DRV_VCC, never +3V3_D.

TERM: POPULATE AT BUS ENDS ONLY

BOOT0 uses existing boot switch/pulldown (PB8_BOOT net name is legacy).
Use HSI/MSI internal clock for USB standalone startup.

VM capacitors: C15 bulk 220uF/63V; C16 at VMA; C17 at VMB; C18 10uF/50V.
Keep VM–PGND loops short; paired pins joined at IC on PCB.

NT1: single PGND–GND copper connection directly beneath IC2 exposed pad.
EP/corner pads to GND; dense thermal vias; 4 layers / 2oz. Motor outputs: 5A routing.

Firmware: DAC1_OUT1 PA4 buffer enabled; reference VDDA/VREF+ (+3V3_A).
Buffered DAC may retain ~0.2V. ENABLE Low or disable DAC for zero current.
Startup: check VM –> set DAC –> set DMODE –> wait >=7.5us –> ENABLE High.
Reset: ENABLE/DMODE/AGC Low; evaluate initially with AGC0=AGC1=Low.

LED_MCU
MCU STATUS (HIGH = ON)

LED_COMM
COMM STATUS (HIGH = ON)

LED_RUN
RUN / ACTIVITY (MCU, HIGH = ON)

LED_FAULT
FAULT (MCU, HIGH = ON)

POWER INPUT 12–24V (J6) / reverse polarity + transient protection

5V BUCK / 400kHz / 1A

EXTERNAL FIRST / USB BACKUP

3.3V MCU SUPPLY

USB only: MCU on, motor VM off.
USB current before enumeration <=100mA total.
VM = ADC/4095 * VDDA * 19; wait >=10ms after power changes.
4.5A is driver absolute peak maximum; Initial VREF limit ~1.97A/phase.
Route high-current PGND via NT1 only; keep BUCK_SW away from ADC.
AP2112 dissipation and motor/Q1 cooling require thermal validation.

USB FS DEVICE / SELF-POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9/_SPI0 VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_M2.USV in firmware.
Internal 3× pull-up / termination. 90–ohm differential PCB pair.

A5600 external sensor / 3.3V I2C
STM32 I2C1: PB6=SCL, PB7=SDA (AF4)
J5: 1=3V3, 2=GND, 3=SCL, 4=SDA
Fit one set of pull-ups on the bus.

VM MEASUREMENT / PAD ADC1_IN5

OSCM: R23=5.1k, C20=270pF; ~67kHz.
Place close to pin35. DRV_VCC: internal 5V, setting circuits only.

VREF: R29=10k series, R30=9.1k to GND, C21=100nF.
3.5V DAC full scale –> VREF 1.579 –> IOUT 1.97A.
R30/C21 GND return near pins29/30, away from OUT/VM.

MOTOR A+/A-/B+/B-

PGND–GND STAR

PGND–GND STAR

Sheet: /	
File: schematic.kicad_sch	
Title:	
Size: A1	Date:
KiCad E.D.A. 10.0.4	Rev:
	Id: 1/1